

Wenyin (Knut) Gu

10431 Adobe Cir, Irvine CA 92617 | knutwygu@gmail.com | +1(650)-272-1546

EDUCATION

- **University of California, Irvine** | 2025 - Present | B.S. in Electrical Engineering (Specialization: DSP)
GPA: 3.655/4.0 | Dean's List (All Quarters)
- **Cañada College** | 2022 - 2025 | A.S. in Engineering
GPA: 4.0/4.0 | Valedictorian, *Summa Cum Laude*, Dean's List (All Semesters)

CONFERENCE PRESENTATIONS

- **“Agentic Affordance: Why LLMs Invite the Intentional Stance” (with Aileen C. Luo)**, *6th International Conference on Philosophy of Mind: Artificial Intelligence (6ICPH)* | 2026.5.
Developed the notion of “agentic affordance” to explain why human-LLM interactions warrant an intentional stance despite a lack of mechanistic interpretability.

PROJECTS

- **Coded Multi-Camera Evidence Recovery under Wi-Fi Jamming** | 2026.7.
Built a Python simulation of a multi-camera home security system under Wi-Fi jamming, using layered Lagrange Coded Computing so that heavy packet loss degrades evidence gracefully instead of erasing it.
Recovered 94% of the evidence versus 36% uncoded, with cross-camera fusion outperforming the best single camera on both accuracy and coverage.
- **Spatial vs. Temporal Edge Detection Simulation** | 2026.3.
Developed Python simulations (NumPy/SciPy) comparing 2D spatial convolution (Sobel operators) with temporal high-pass filtering (difference equations) for edge detection.
Demonstrated event-based temporal filtering achieves superior signal sparsity and efficiency in motion tracking by mathematically suppressing static backgrounds to zero.
- **Music Reactive LED System** | 2026.3.
Designed a custom microphone preamplifier and 3-band active analog crossover (LM358 Op-Amps) to hardware-separate audio frequencies (Bass, Midrange, Treble).
Programmed an Arduino non-blocking state machine and an interactive OLED GUI to process multi-channel ADC signals into three real-time LED visualization modes.

EXPERIENCE

- **IEEE-Eta Kappa Nu (HKN) Webmaster** | *UC Irvine* | 2026.5. - Present
Maintain and architect the official honor society website, managing regular data updates for event schedules, membership rosters, and technical archives to streamline chapter communications.
- **Legacy Robotics (Communications Subsystem Engineer)** | *UC Irvine* | 2026.4. - Present
Audit the rover's 5 GHz video and 900 MHz MAVLink telemetry links for the University Rover Challenge and design a staged upgrade: feedline and rover-antenna fixes worth 6-10 dB of link margin, a GPS- and RSSI-guided auto-tracking base-station antenna (ESP32 and stepper pan-tilt), and link-aware adaptive video with dual-link failover.
- **Tutor of Mathematics** | *Cañada College* | 2022 - 2024
Provided one-on-one, drop-in tutoring at the Learning Centre, guiding students through mathematical problems and supporting their development of analytical skills.
- **Fundamentals of Data Science Workshop** | *Stanford University* | Summer 2023
Completed a 4-week intensive program encompassing Applications of Data Science, Python, Statistics, Mathematical Optimization, Linear Algebra, and Responsible AI.
- **Outstanding Service Contributor (Water Supply Team Lead)** | *Shanghai World Expo* | 2010
Led a team to conduct comprehensive safety inspections of water supply networks and enforce administrative penalties for water-related violations, successfully safeguarding the drinking water security for over 73 million visitors throughout the 6-month event.

SKILLS

- **Software & Programming:** Python, MATLAB, SolidWorks, Onshape, C++, Mission Planner.
- **Hardware & Electronics:** Analog Circuit Design (Op-Amps, Active Filters), RF Link Budgeting & Antenna Selection, Microcontroller Integration (ADC, I2C), Embedded UI (OLED, Rotary Encoder).

AWARDS and HONOURS

- IEEE-Eta Kappa Nu (HKN) | 2026
- Phi Theta Kappa | 2023
- Dean's List | 2022 - Present